Daily Research Digest - 2026-04-21

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter

Synthesis

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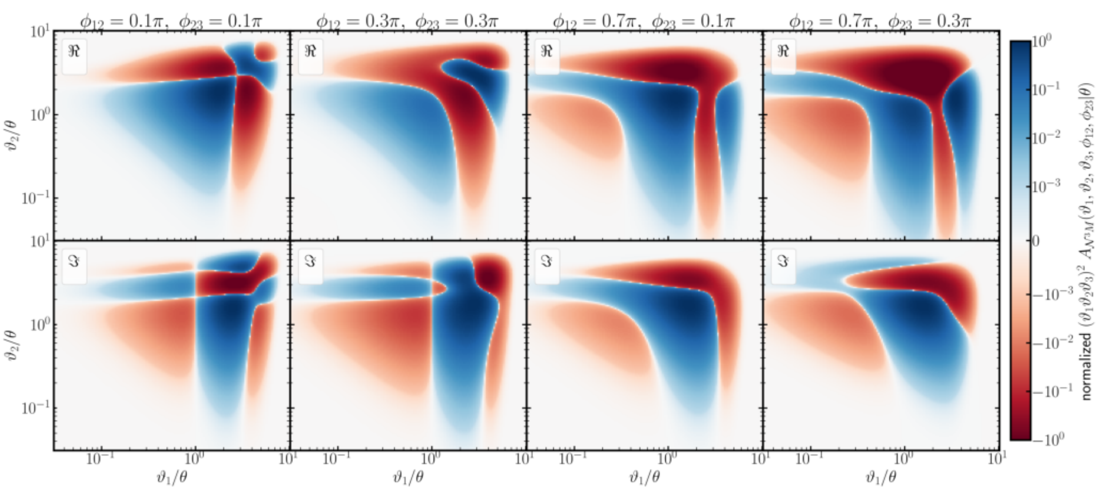
Paper Summaries and Figures

1. Fourth-order galaxy-galaxy-lensing: Theoretical framework and direct estimation

Focus: gravitational lensing
Authors: Jonathan Oel, Lucas Porth, Peter Schneider, Elena Silvestre-Rosello
Published: 2026-04-20T15:04:01+00:00
Summary: Traditional galaxy-galaxy lensing is a well-established method of probing the statistical properties of the Universe's matter and galaxy distribution. However, this measure does not carry all the statistical information, provided the matter and galaxy distribution contain non-Gaussian features. In order to study these non-Gaussianities, it is necessary to consider higher-order statistical measures. The aim of this work is to extend the analytical basis describing the statistical correlations between galaxies and shear to the fourth order, with special emphasis on the associated aperture statistics. In order to include fourth-order statistics in future analysis of the relation between mass and galaxies, we further investigate whether we can expect to detect these statistics from observations of stage IV surveys. We define the four-point correlation function (4PCF) between the shear and the positions of triplets of foreground galaxies and derive its relation to the respective trispectrum. We convert the 4PCF to aperture statistics and derive the analytical form of the respective filter function, which we then implement in a numerical integration pipeline. Furthermore, we develop a direct estimator that allows us to measure galaxy-mass aperture moments of arbitrary order on pixelized data using a Fast-Fourier-Transform (FFT) algorithm. We show that the corresponding aperture measure ^3 M_ap can be calculated with sub-percent accuracy on relevant aperture scales, θ, by means of numerical integration. Furthermore, we apply the FFT-based direct estimator to a mock catalog with a realistic stage IV survey setup on a sky area of 2000~deg^2, and detect the connected part of the aperture statistics ^3 M_ap(θ) with a signal-to-noise ratio of roughly nine on small aperture scales.

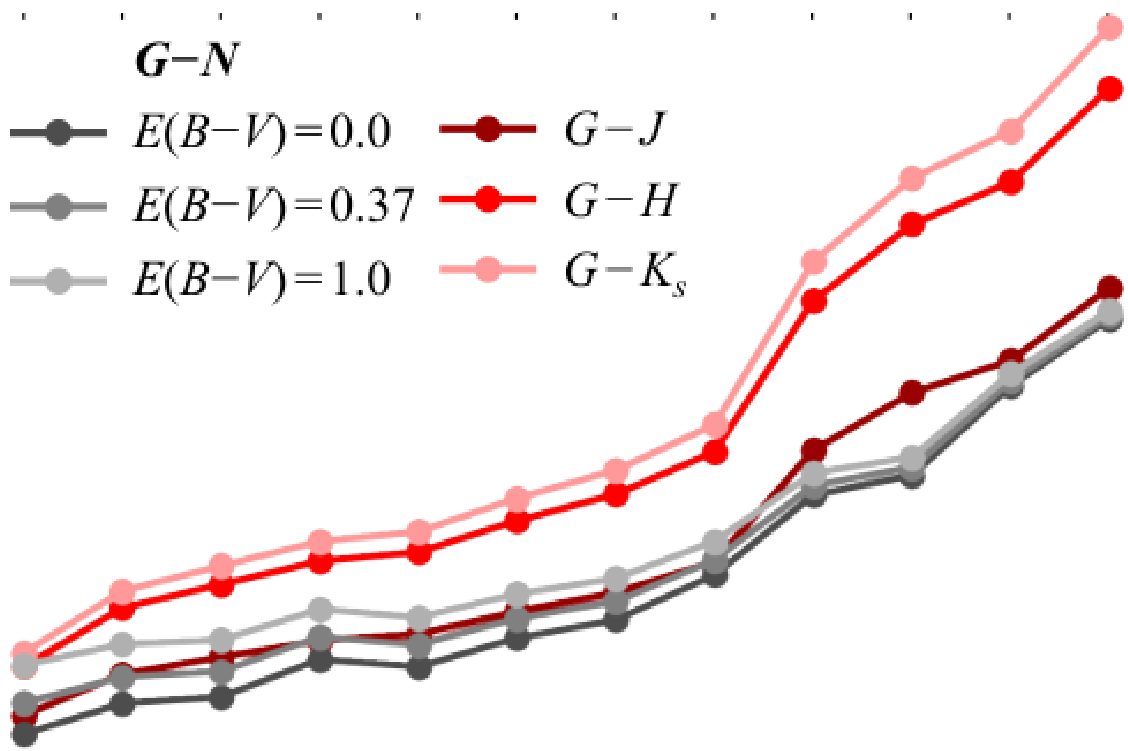
Selected figures



2. From Gaia to GaiaNIR: I. Probing dark matter halos in globular clusters

Focus: dark matter
Authors: I. Henum, D. Hobbs, Ó. Jiménez-Arranz, P. J. McMillan, R. P. Church
Published: 2026-04-20T13:38:13+00:00
Summary: The proposed GaiaNIR mission would extend Gaia's astrometric capabilities into the near-infrared, improving astrometric precision and enabling observations in heavily dust-obscured regions. In this work, we investigate the impact of GaiaNIR on the detectability of dark matter halos in globular clusters by comparing its performance with that of Gaia. Expected observations from future Gaia data releases and GaiaNIR are modeled for a globular cluster with properties similar to M4. The cluster is simulated with a range of dark matter halo sizes and varying levels of extinction, allowing a direct assessment of each mission's ability to detect and distinguish dark matter halos across different extinction conditions. To support this comparison, the relationship between the Gaia G band and several near-infrared bands is examined. We find that Gaia can resolve extended dark matter halos in low-extinction conditions, but its performance degrades significantly as extinction increases. In contrast, GaiaNIR reduces statistical uncertainties and retains sensitivity to extended halos even with strong extinction. These results indicate that GaiaNIR would both strengthen constraints on kinematic signatures accessible to Gaia and enable the study of clusters in heavily obscured regions that are currently beyond Gaia's reach.

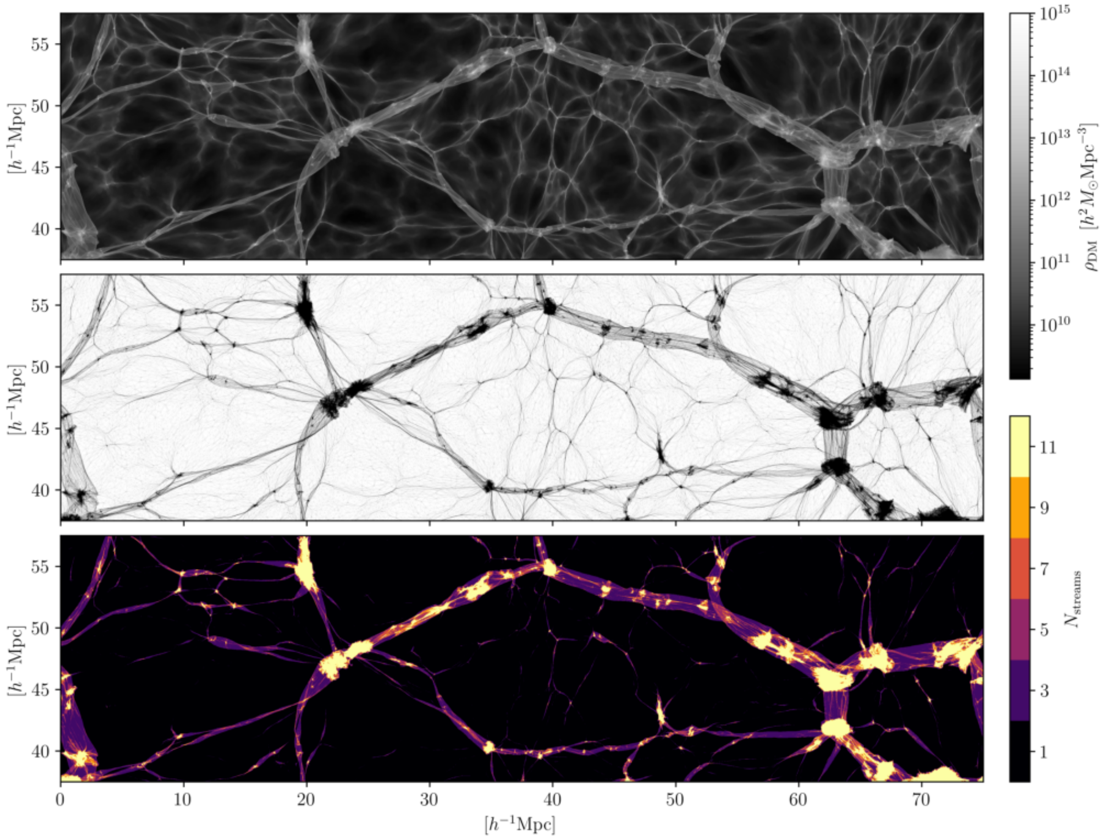
Selected figures



3. Galaxy Populations in the IllustrisTNG Caustic Skeleton

Focus: dark matter
Authors: Benjamin Hertzsch, Job Feldbrugge, Rien van de Weygaert
Published: 2026-04-20T12:56:25+00:00
Summary: The caustic skeleton is a parameter-free and mathematically rigorous formalism for tracing the hierarchical formation history of the multiscale cosmic web from the singularities in the underlying dark matter flow. In the present study, we explicitly use the multistreaming nature of the cosmic mass distribution to address the influence of the weblike embedding on the galaxy populations and discern their properties in different web environments. To this end, we construct the multiscale caustic skeleton of the dark mass distribution in the state-of-the-art suite of the large-scale IllustrisTNG simulations. In addition to the multistreaming dark matter density field, we assess the characteristic properties of the intergalactic baryonic gas in the vicinity of the caustics. Next, we associate the galaxies with the voids, walls, filaments and cluster nodes, and investigate their colours and star formation activities. A unique feature of the analysis is that it explicitly addresses the multiscale aspects with respect to the galaxy population, assessing issues such as the fraction of (blue) galaxies as a function of the scale of the cosmic web pattern and its caustic features. We find that the galaxy properties form a continuum in the scale-space cosmic web. Intimately coupled to the hierarchical build-up of the cosmic structure, it also allows us to systematically assess the impact of the formation time of the various structural components of the cosmic web on the galaxy properties. This furthers insight into the establishment of the observed colour-density relation of galaxies.

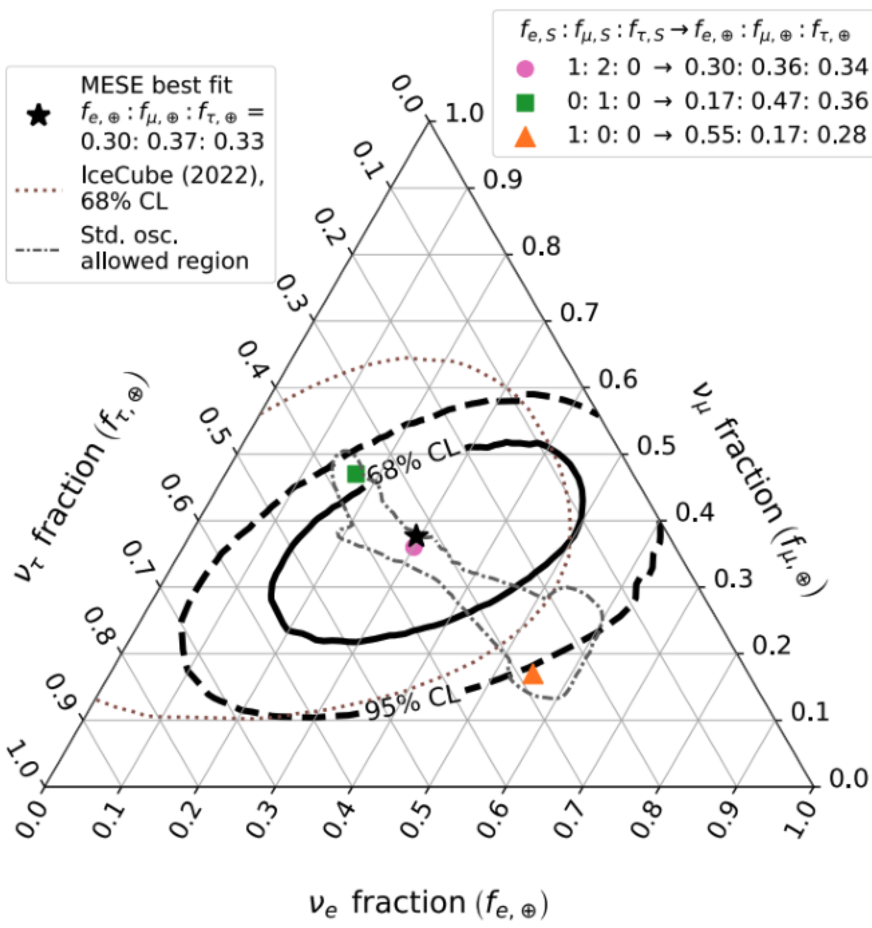
Selected figures



4. Latest results from the IceCube Neutrino Observatory

Focus: dark matter
Authors: Thijs Juan van Eeden
Published: 2026-04-20T12:34:25+00:00
Summary: The IceCube Neutrino Observatory has opened a new window into the high-energy Universe, providing measurements of neutrinos over a broad energy range. This contribution presents recent results, including a follow-up on the first identification of a steady neutrino source NGC 1068, measurements of the flavor composition of the diffuse astrophysical flux, limits on prompt atmospheric neutrinos, and searches for neutrinos from dark matter annihilation in the Sun. These measurements probe neutrino production mechanisms, fundamental particle interactions, and physics beyond the Standard Model. Looking forward, the recently deployed IceCube Upgrade will enhance sensitivity to lower-energy neutrinos and reduce systematic uncertainties, while the planned IceCube-Gen2 will expand the detector volume, increase the neutrino detection rate, and extend energy reach, enabling more detailed studies of cosmic sources and high-energy particle physics.

Selected figures



5. In-depth analysis of the clustering of dark matter particles around primordial black holes. Part III: CMB constraints

Focus: dark matter
Authors: Julien Lavalle, Vivian Poulin, Pierre Salati
Published: 2026-04-20T09:30:49+00:00
Summary: In a mixed dark matter scenario in which primordial black holes (PBHs) would co-exist with thermally produced self-annihilating particles, one expects the former to be surrounded by extremely dense halos made of the latter, built up during radiation domination. Here, as a continuation of previous work, we derive observational limits on such a scenario from a full statistical analysis of cosmic microwave background (CMB) data. We quantify how a tiny fraction of PBHs could restrict the parameter space available to thermal particle dark matter, limiting the s-wave annihilation cross section to values 10^-30\, cm^3/s\,(/100\, GeV)\,(/10^-6)^-3 if PBHs are typically heavier than 10^-10\,, which can also be turned into constraints on PBHs in this mass range. In contrast, asteroid mass or lighter PBHs could live in perfect peace with these particles. Finally, we shortly discuss the implications of the recent tentative interpretation of Subaru-HSC microlensing events as PBHs.

6. Efficiently emulating distribution functions in gigaparsec volumes for varying cosmological parameters

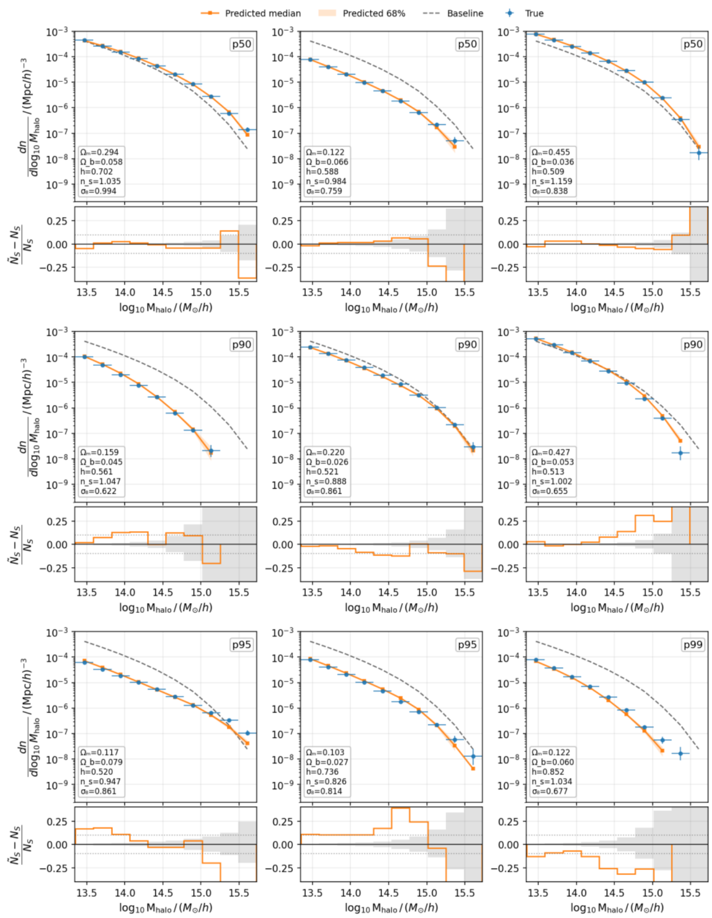
Focus: dark matter

Authors: Christopher C. Lovell, Max E. Lee, William J. Roper, Daniel Anglés-Alcázar, Shy Genel, Shivam Pandey, Francisco Villaescusa-Navarro

Published: 2026-04-20T09:02:04+00:00

Summary: We present a new method for emulating the halo mass function (HMF) and other distribution functions in large effective volumes, down to low halo masses, whilst simultaneously modifying large ranges of parameters, for a fraction of the cost of traditional periodic cosmological simulations. We demonstrate the method by selecting small regions, V (50 \,h^-1 Mpc)^3, with a range of overdensities from the Quijote suite, consisting of tens of thousands of (1 \,h^-1 Gpc)^3 N-body simulation volumes run with varying ΛCDM parameters. We train a differentiable emulator, conditioned on the overdensity of the region and these global parameters, to reproduce the halo mass function in these regions. We then successfully recover the global distribution of halo masses of the entire box by integrating over the overdensity distribution. Our approach uses just \,0.026% of the original simulation volume, and suggests that suites of targeted zoom' simulations, extracted from low resolution parent volumes, can be used to emulate large volume simulations at a fraction of the computational cost, whilst simultaneously pushing the dynamic range to much lower masses than can be achieved in periodic simulations. We discuss emulation of other key dark matter and baryonic distribution functions, as well as higher order statistics, with implications for the interpretation of upcoming wide field surveys on observatories such as Euclid, Roman and Rubin.

Selected figures



Predicted median
Predicted 68%
Baseline
True
p50
p50
p50
p90
p90
p90
p95
p95
p99
Ωₘ=0.294 Ω_b=0.058 h=0.702 n_s=1.035 σ₈=0.994
Ωₘ=0.122 Ω_b=0.066 h=0.588 n_s=0.984 σ₈=0.759
Ωₘ=0.455 Ω_b=0.036 h=0.509 n_s=1.159 σ₈=0.838
Ωₘ=0.159 Ω_b=0.045 h=0.561 n_s=1.047 σ₈=0.622
Ωₘ=0.220 Ω_b=0.026 h=0.521 n_s=0.888 σ₈=0.861
Ωₘ=0.427 Ω_b=0.053 h=0.513 n_s=1.002 σ₈=0.655
Ωₘ=0.117 Ω_b=0.079 h=0.520 n_s=0.947 σ₈=0.861
Ωₘ=0.103 Ω_b=0.027 h=0.736 n_s=0.826 σ₈=0.814
Ωₘ=0.122 Ω_b=0.060 h=0.852 n_s=1.034 σ₈=0.677
$\frac{dn}{d\log_{10} M_{\rm halo}}$ / (Mpc/h)$^{-3}$
$\frac{\tilde{N}_S - N_S}{N_S}$
$\log_{10} \mathrm{M_{halo}}$ / ($M_\odot/h$)

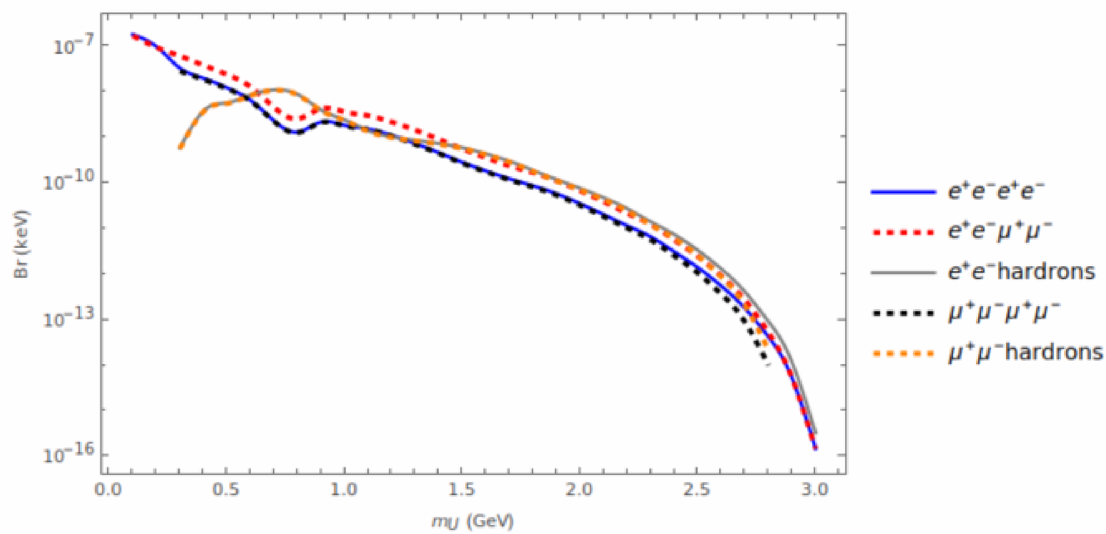
7. Probing Cosmic-Ray-Boosted and Supernova-Sourced Sub-GeV Dark Matter with Paleo-Detectors

Focus: dark matter
Authors: Xiaoyong Chu, Yue-Lin Sming Tsai, Mei-Wen Yang
Published: 2026-04-20T06:30:15+00:00
Summary: Astrophysical dark matter particles with masses well below GeV-scale can be difficult to detect using conventional nuclear recoil experiments due to their low velocities in our Milky Way halo. Elastic scattering with high-energy cosmic rays or thermal production inside core-collapse supernovae can accelerate sub-GeV DM to (semi-)relativistic velocities, producing nuclear recoil energies above the keV threshold that paleo-detectors can record over geological timescales. Using olivine as the target with 100\,g exposure, we compute track length distributions from such (semi-)relativistic dark matter fluxes, incorporating all major backgrounds (neutrinos, uranium-chain neutrons, thorium recoils) with a statistical analysis on an Asimov dataset. We derive 95 C.L. projected sensitivity of paleo-detectors to the DM-nucleon cross section for dark matter masses between a few MeV and hundreds of MeV. Our results show that paleo-detectors are able to probe large parameter regions that are not covered by current and near-future experiments designed to detect dark matter and neutrinos. In particular, paleo-detectors offer a unique ability to record the dark matter flux from Galactic supernova events over geological times. Such cumulative exposure enables sensitivity gains of a few orders of magnitude compared to conventional experiments.

8. Searching for dark photons in J/ψ decays

Focus: dark matter
Authors: Xiao Liang, Chun-Yuan Li, Bin-Peng Shang, Zong-Guo Si, Hong-Xin Wang, Xing-Hua Yang, Dai-Xing Zhang
Published: 2026-04-20T02:21:28+00:00
Summary: A dark photon is an Abelian gauge boson from a new U(1)_D gauge symmetry, coupled to the Standard Model via kinetic mixing, with inducing an effective coupling to the electromagnetic current and g_χ to a stable dark matter particle χ. We study J/ψ two-body and four-body decays via a light-mass dark photon (m_U < 3.0 GeV) in the framework of non-relativistic QCD (NRQCD), considering both visible and invisible decays of the dark photon into SM fermions or dark sector particles. Numerical results for the decay ratios Γ/Γ_J/ψ and expected event numbers at BESIII are presented, along with the significance S/B and p_T distributions where applicable. Our results show that, for m_U < 2m_χ, the two-body final state decay channels of the J/ψ mediated by a dark photon have event yields 0 37 with significances of 10^-510^-6, while the four-body final state channel yields about 94172 events in the very low mass region m_U < 0.2\ GeV. For m_U 2m_χ, the invisible two-body final state decay channel yields 012 events with better significance 10^-110^-3, and the invisible four-body final state decay channel yields 0129 events, whereas visible decays are all severely suppressed.

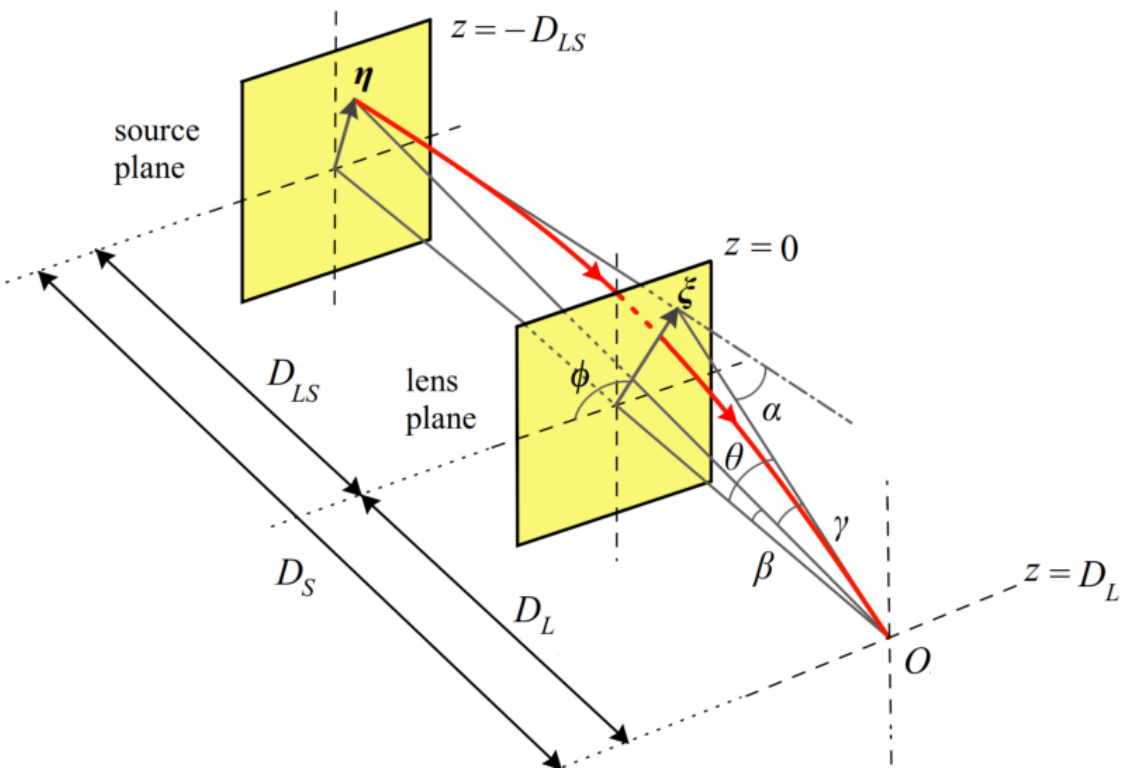
Selected figures



9. Weak Gravitational Lensing: A Brief Overview

Focus: gravitational lensing
Authors: Partha Pratim Basumallick, Saheb Das, Bhaswati Mandal, Subhadip Sau
Published: 2026-04-19T10:01:12+00:00
Summary: Gravitational lensing constitutes one of the most direct observational manifestations of spacetime curvature and provides a powerful probe of compact astrophysical objects. In this work, we present a comprehensive analysis of the bending of light in curved spacetime, beginning with the fundamental aspects of gravitational lensing and the Newtonian approximation to light deflection. The relativistic formulation of lensing is then developed through the lens equation, lensing potential, and a geometrical interpretation of Fermat's principle using an effective refractive index in curved spacetime. Photon trajectories and light deflection are subsequently investigated in static, spherically symmetric geometries, followed by a detailed study of photon motion in the equatorial plane of the Kerr spacetime. Analytical expressions for the closest approach distance and critical parameters governing photon orbits are derived. Furthermore, the bending angle is examined using the Rindler-Ishak method and the Gauss-Bonnet theorem within the optical geometry. Finally, the analysis is extended to axisymmetric spacetimes using the OIA and GW-OIA formalisms, providing a unified geometrical framework for computing light deflection in both static and rotating gravitational fields.

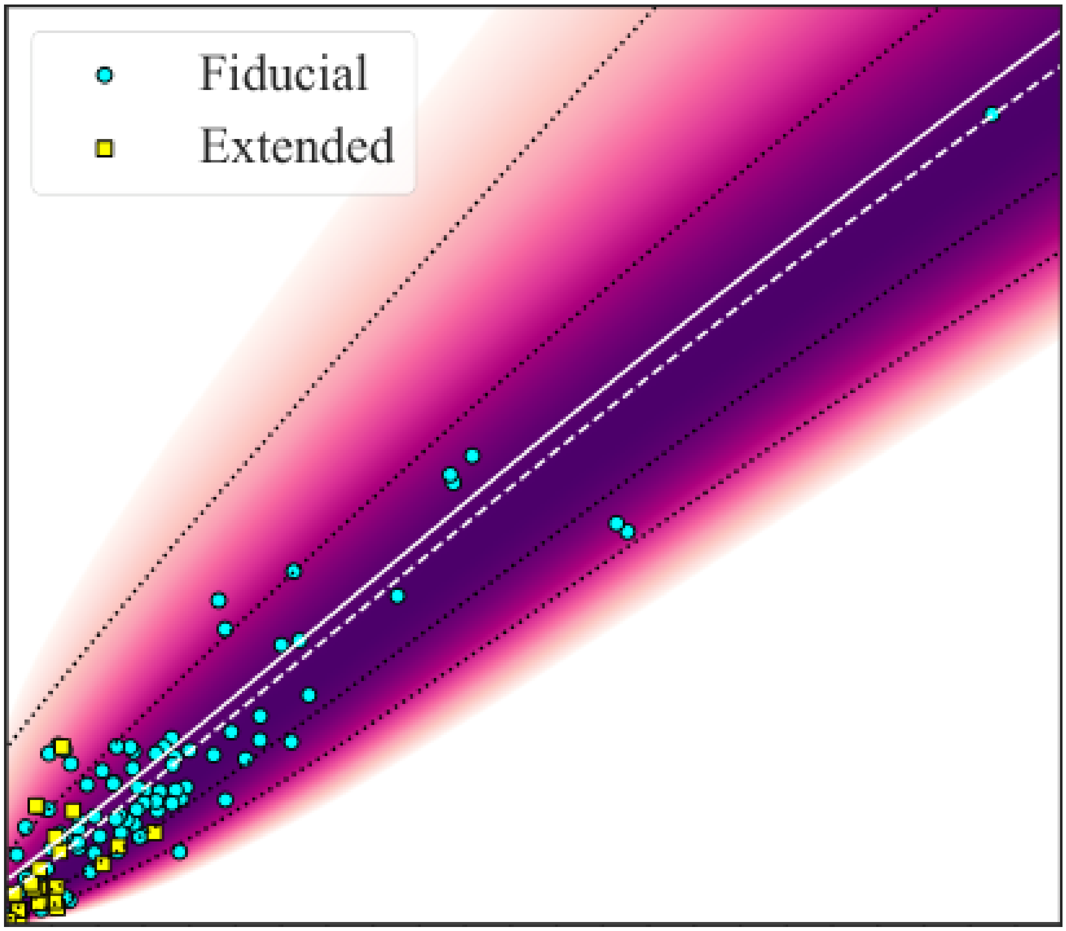
Selected figures



10. Signatures of Suppressed Matter Clustering revealed by Fast Radio Bursts

Focus: gravitational lensing
Authors: Kritti Sharma, Elisabeth Krause, Vikram Ravi, Liam Connor, Dhayaa Anbajagane, Pranjal R. S
Published: 2026-04-18T22:37:05+00:00
Summary: Complex astrophysical processes regulate the growth of galaxies by injecting energy and momentum into their surroundings, redistributing baryons across megaparsec scales. The clustering of matter on these scales, as measured via weak lensing and galaxy surveys, encodes critical cosmological information on the dynamical dark energy, the nature of dark matter and the sum of neutrino masses. The suppression of matter clustering due to feedback processes limits the interpretation of cosmological measurements. Multiple probes of the baryon distribution have attempted to quantify the strength of feedback via measurements of suppression in the matter power spectrum. The dispersion measures (DMs) of fast radio bursts (FRBs) have emerged as a powerful new probe of baryons, with the advantage over other probes of being unbiased with respect to density and temperature. Here, we use a sample of 109 FRBs with redshifts and DMs to directly measure the spatial fluctuations in the baryon density field, quantifying the effects of feedback on the matter power spectrum at scales of k 0.1-3 h/Mpc, and the gas fraction in galaxy groups and clusters (10^13-10^15 M_). We use a halo-model prescription to conduct inference, and find that FRB data reduces the posterior variance at k 1 h/Mpc by a factor of 8 relative to the prior. The statistical precision of inferred FRB constraints is similar to other baryon tracers, while probing a complementary redshift regime (z 0.3). A comparison with several hydrodynamical simulations excludes extreme large-scale feedback scenarios at 2σ confidence. This work establishes FRBs as a sensitive probe of feedback-regulated structure formation. As next-generation experiments deliver orders-of-magnitude larger samples, FRBs are poised to drive the constraints on baryonic physics in the era of precision cosmology.

Selected figures


Fiducial
Extended